

INSTRUCTION MANUAL

Simrad AD50
Analog Drive

Note!

Simrad AS makes every effort to ensure that the information contained within this document is correct. However, our equipment is continuously being improved and updated, so we cannot assume liability for any errors which may occur.

Warning!

The equipment to which this manual applies must only be used for the purpose for which it was designed. Improper use or maintenance may cause damage to the equipment or injury to personnel. The user must be familiar with the contents of the appropriate manuals before attempting to operate or work on the equipment.

Simrad AS disclaims any responsibility for damage or injury caused by improper installation, use or maintenance of the equipment.

Copyright

© 2003 Simrad AS

The information contained within this document remains the sole property of Simrad AS.

No part of this document may be copied or reproduced in any form or by any means, and the information contained within is not to be communicated to a third party, without the prior written consent of Simrad AS.

Instruction Manual

This manual is intended as a reference guide for installing the AD50 Analog Drive and how the AD50 is used to interface an analog rudder system to a Simrad AP50 autopilot system. The manual is intended to be used in conjunction with the AP50 autopilot manual for software setup.

Please take time to read the manuals to get a thorough understanding of how to interface an analog or proportional rudder control system to an autopilot.

Document revisions

Rev.	Date	Written by	Checked by	Approved by
A	180603	NG	<i>G.K.</i>	<i>V.P.</i>
B				
C				
D				

Document history

Rev. A Original issue

Table of contents

1	INTRODUCTION.....	5
2	TECHNICAL SPECIFICATIONS.....	6
3	INSTALLATION	7
3.1	Mechanical Mounting	7
3.2	Electrical connection	7
	Ground connection	7
	Continuous voltage (± 10 VDC).....	8
	Proportional rudder speed command with directional switching	11
	Danfoss PVEM valve	12
	Directional Solenoids (closing contact).....	13
	Drive enable.....	14
	Software Setup.....	15
4	MAINTENANCE	17
4.1	Exchange of software program.....	17
5	TROUBLESHOOTING	19
5.1	Failure Mode Effect Analyses	20
6	SPARE PARTS LIST	21

This page is intentionally left blank.

1 INTRODUCTION

The AD50 Analog Drive is designed to provide analog or proportional control of a rudder in an AP50 system by either continuous voltage, or Danfoss PVEM valve signal.

The property of an analog signal is that the rudder angle is defined as a voltage, normally within $\pm 10V$ where 0V defines midship rudder position and the $\pm 10V$ defines ± 45 degrees hard-over position.

Note !

No rudder feedback unit is required for this option.

The proportional signal is different in the way that the polarity of the $\pm 10V$ signal determines the direction of the rudder movement, while the voltage amplitude determines the rudder speed.

Note !

Rudder feedback unit is required for this option.

The Danfoss PVEM valves require a special designated offset signal for operation.

The analog rudder output signal is calculated in the AD50, based on operational mode and rudder command received over the RobNet network from other system units. Set-up from the control unit and any errors in the analog rudder interface is communicated via the RobNet. All settings in the AP50 Control Unit for rudder control signal are stored in the AD50.

2 TECHNICAL SPECIFICATIONS

Dimensions:	See Figure 2-1.
Weight:	0,8 kg (1.8 lbs)
Material:	Epoxy coated aluminium
Environmental Protection:	IP44
Supply and interface:	RobNet, 2 connectors
Cable inlets:	Rubber glands for cable diam. 10-14 mm
Cable length:	7 m (23 ft.)
Mounting:	Bulkhead mount
Compass safe distance:	0.2 m (0.7 ft)
Temperature range:	
Operation:	-25 to +55°C (-13 to +130°F)
Storage:	-30 to +80°C (-22 to +176°F)
Proportional control:	
Danfoss PVEM:	Nominal UDC=12/24V, I=0.25/0.5mA, neutral 0.5*Un, control range 0.25*UDC to 0.75*UDC, valve saturated for <0.25*UDC or >0.75*UDC.
Analog angle or proportional control:	
Internal supply:	Control range ±10V, max. 5 mA, galvanic isolated
External supply:	UDC 12-24VDC, control range 0- UDC or ±UDC/2, max. 5 mA
Port/stbd directional signal:	On/off, open collector, galvanic isolated, plus or minus, 3A max.
Rudder enable output:	Open collector, external or internal +, max 500 mA The internal +12V output is limited to 100mA and may be used for an external relay operated by Hi/Lo output to switch the rudder control signal between autopilot and external manual control

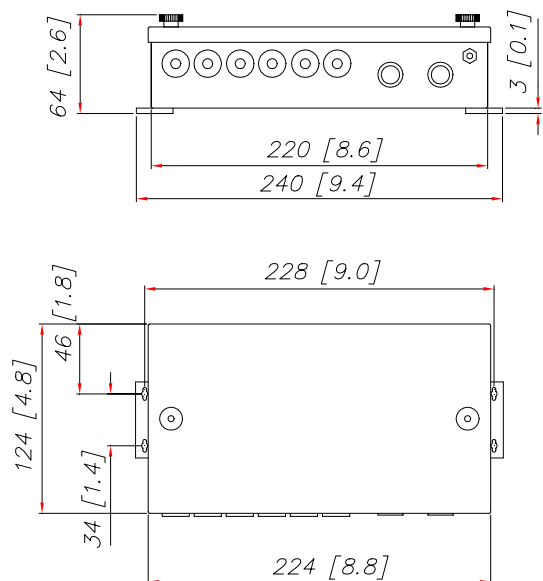


Figure 2-1 AD50 Dimensions

3 INSTALLATION

3.1 Mechanical Mounting

The AD50 is normally installed inside a console or locker close to a RobNet unit and within the length of the supplied 7 m (23') RobNet cable. The unit does not have controls that need to be adjusted during installation or use. It should be installed with the cable inlet and the RobNet connectors facing down for best internal heat dissipation. The AD50 is designed to operate in a location that provides ambient temperatures below +55°C (+130°F). It is fastened to the panel/bulkhead by the external mounting brackets.

Note !

AD50 is not weatherproof, and must be installed in a dry location!

3.2 Electrical connection

The AD50 can be connected in several ways to a rudder system. The connections are as follows:

1. Continuous voltage
 - Analog rudder angle order (TB3) ($\pm 10\text{VDC}$)
or
 - Proportional rudder speed command (TB3) with directional switching (TB1) ($0\text{--}\pm 10\text{VDC}$)
2. Danfoss valve
 - Direct proportional command (TB4)
3. Rudder enable signal (TB2)

Ground connection

To make a good ground connection, strip about 1 cm (0.4") off the cable's insulation and pull the screen backwards to cover the insulation. Position the straps as shown in Figure 3-1. Tighten well to make sure the screen makes good contact.

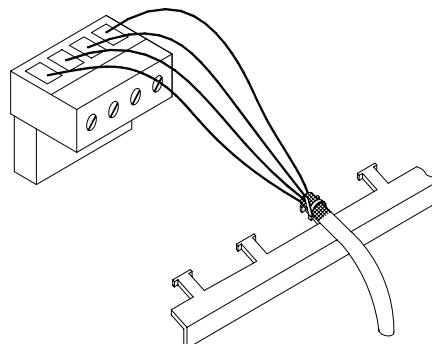


Figure 3-1 Grounding

Continuous voltage (± 10 VDC)

AD50 has several possibilities for analog or proportional rudder control by a continuous signal.

A nominal 0 ± 10 V control signal at TB3 no. 1-2 shown in Figure 3-2. Note that this voltage is 0 ± 10 V regardless of the operating voltage for the autopilot.

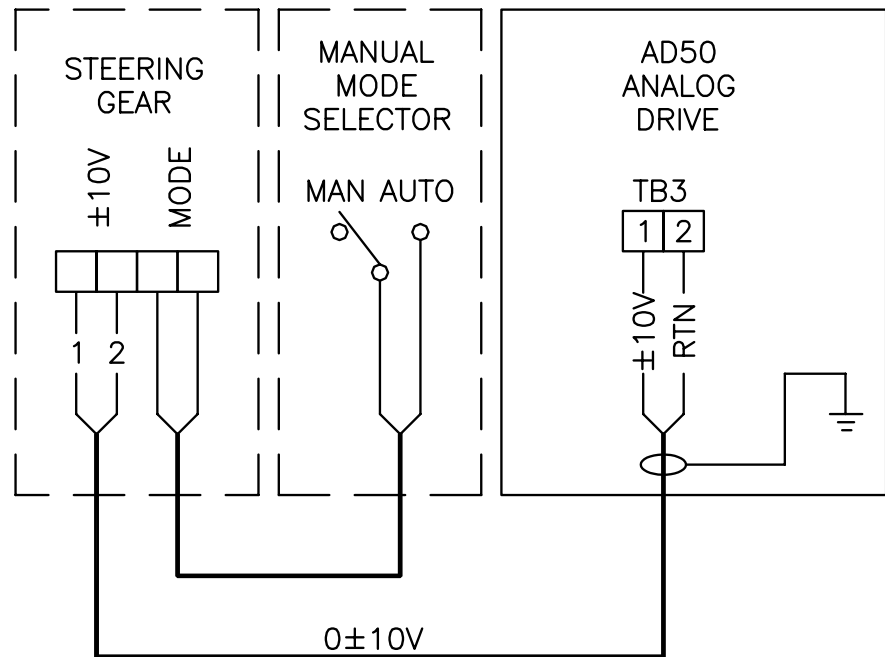


Figure 3-2 Independent ± 10 V output

A control signal at TB4 that is dependent upon the externally supplied voltage (12-24 VDC) shown in Figure 3-3. If the supplied voltage is 24 VDC, the default output will be 0 ± 12 V, using TB4-2 (RTN) as reference. This reference is half of the supply voltage via a voltage divider.

If the supplied voltage is 12 V, only 0 ± 6 V can be obtained. Therefore alternative 1 (Figure 3-2) must be used for ± 10 V signal and 12V mains supply.

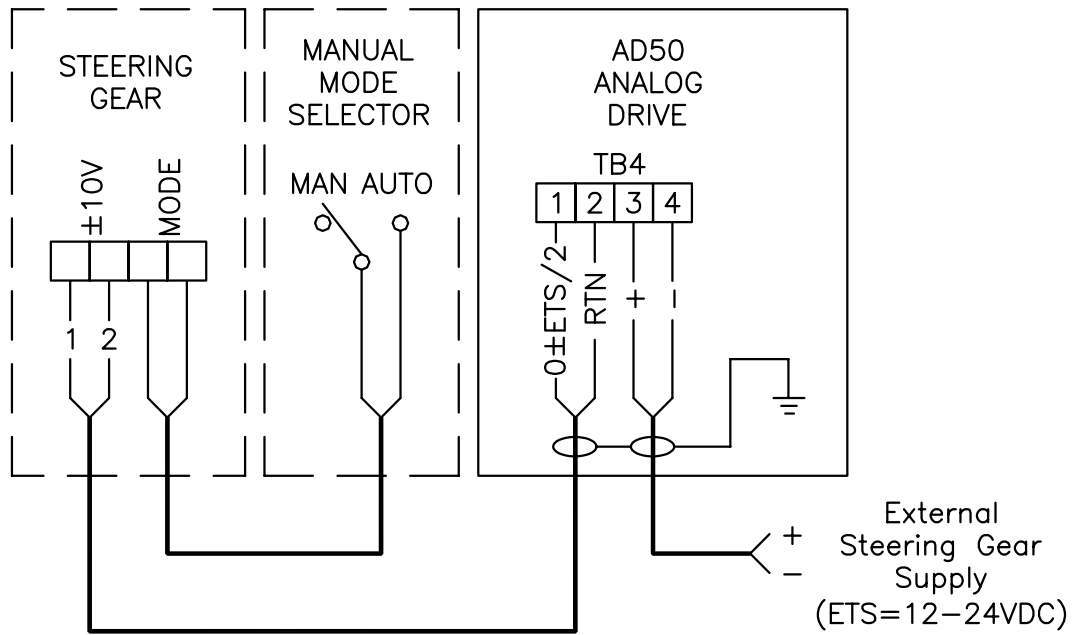


Figure 3-3 Dependent ± 10 V output

If the supplied voltage return is used as reference, the default output will be 12V ($24\text{V}/2$) $\pm 12\text{ V}$ or 6 V ($12\text{V}/2$) $\pm 6\text{ V}$.

The rudder direction, amplitude, and offset adjustments are made via the software menu in the autopilot (refer to “Software Setup” on page 15).

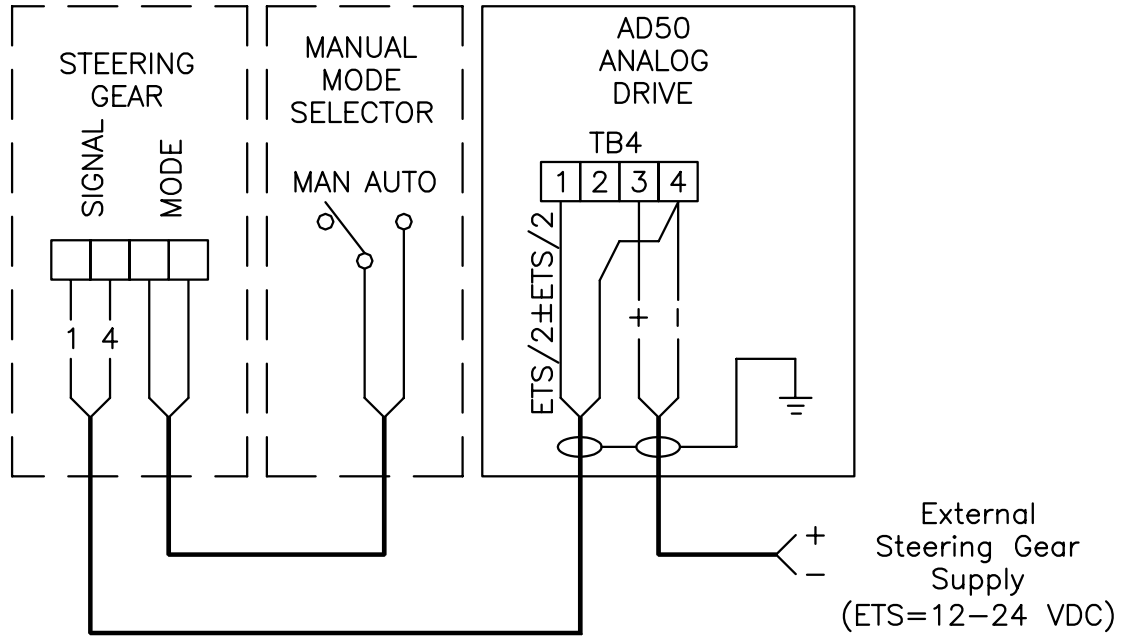


Figure 3-4 Positive variable output signal

Note !

The Manual Mode Selector is optional and not supplied by Simrad (See also Drive enable, 14).

Proportional rudder speed command with directional switching

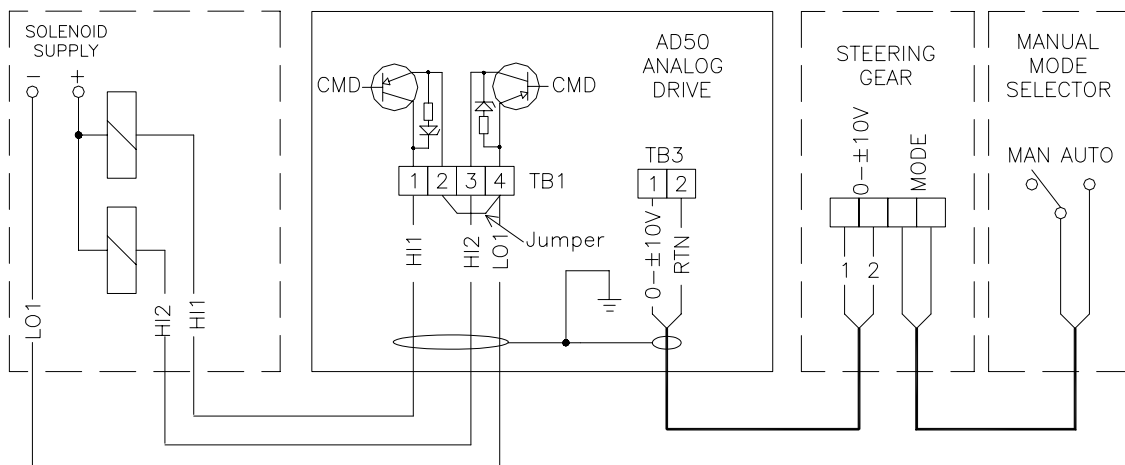


Figure 3-5 Proportional rudder speed command with directional switching - common positive supply

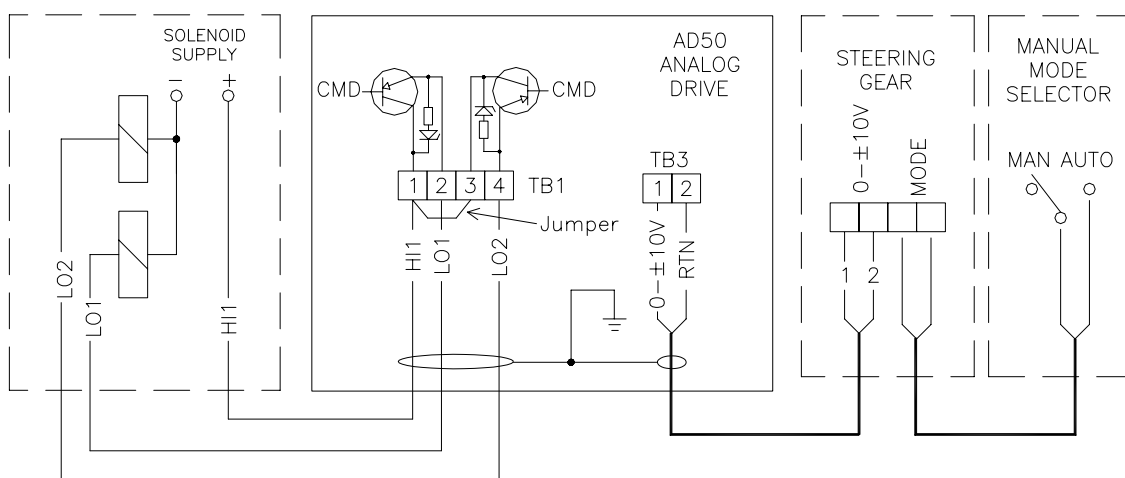


Figure 3-6 Proportional rudder speed command with directional switching - common negative supply

Danfoss PVEM valve

The connections are made as shown on Figure 3-7. The UDC+ and UDC- are used to power part of the AD50 PCB to obtain galvanic isolation between the autopilot and valve supply.

Caution

Do not connect other external power to TB4 other than UDC+ and UDC- supplied from the Danfoss PVEM valve.

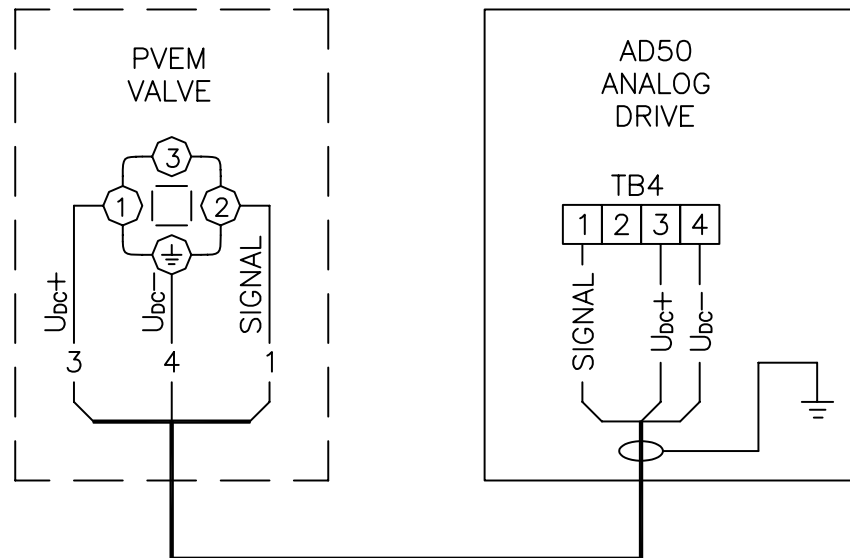


Figure 3-7 Connections to Danfoss PVEM valve

Directional Solenoids (closing contact)

Some steering gear may need a directional on/off signal in addition to the continuous control.

For this purpose the AD50 has a closing contact output. The solenoids can be connected for common positive or common negative voltage by a 3-conductor cable as shown in Figure 3-8 and Figure 3-9.

Normally the solenoids can be operated directly from the boats manual system and the autopilot. If not, refer to “Drive enable”, page 14 for switch arrangement.

Note !

Make sure the connections are made with correct polarity. Else the EMF spike protection diodes across the output transistors may be damaged.

Maximum current: 3 Amps.

For software setup, refer to the AP50 manual.

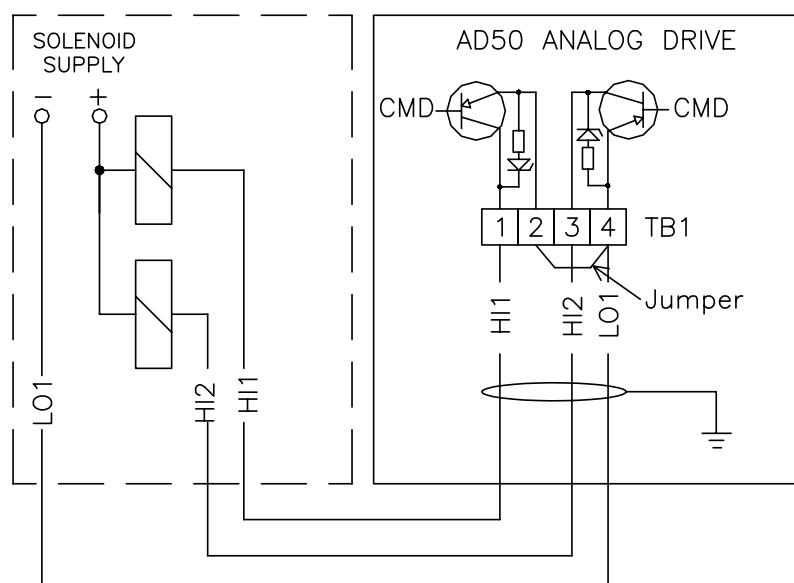


Figure 3-8 Connection to Directional solenoids, common positive supply

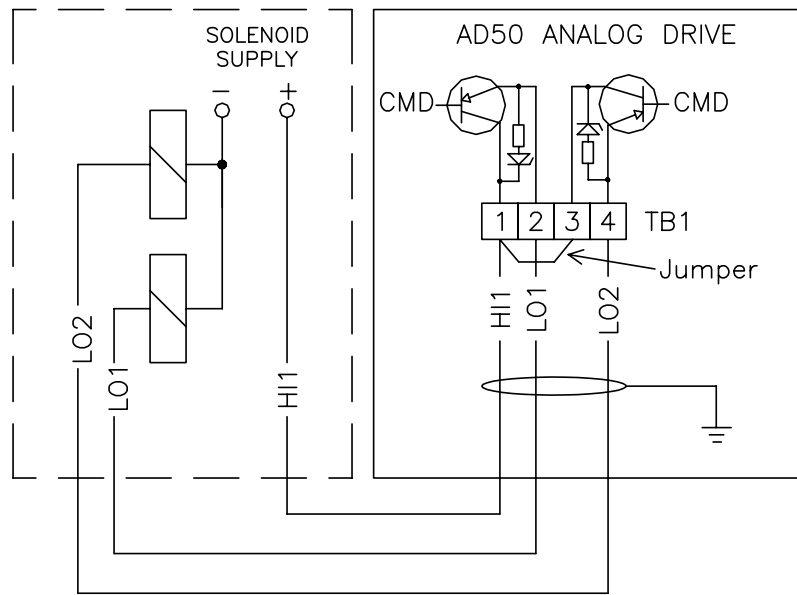


Figure 3-9 Connection to Directional solenoids, common negative supply

Drive enable

If the rudder shall be operated both by the boats manual control handle and by the autopilot, an external closing contact to enable the autopilot control signal may be required. A manual mode selector for this is suggested in Figure 3-2, Figure 3-3 and Figure 3-4.

AD50 also outputs a galvanic isolated drive enable signal that can be used to automatically switch between the manual rudder control and the autopilot. This is shown in Figure 3-10 and Figure 3-11.

The transistor Q105 is switched on, energizing the relay, whenever the autopilot is controlling the rudder in NFU by the rudder commands, or when in FU, AUTO and NAV mode.

Note !

NFU-mode is not available for analog rudder control.

In Figure 3-10, a 12V relay is used with supply voltage from AD50. The jumper switch S1 on the AD50 PCB must be set to position A-B. Maximum relay coil current is limited to 100 mA.

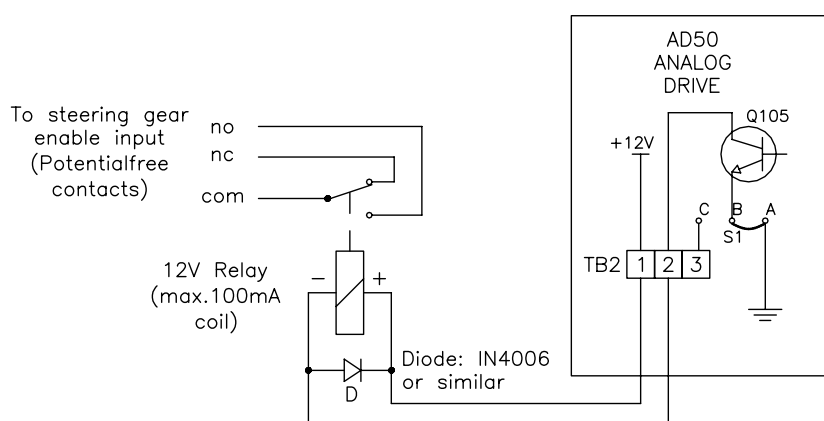


Figure 3-10 Rudder Enable, Alternative 1

In Figure 3-11, a 12-24-32V externally powered relay is used. The jumper switch S1 on the AD50 PCB must be set to position B-C. Maximum relay coil current is limited to 500mA.

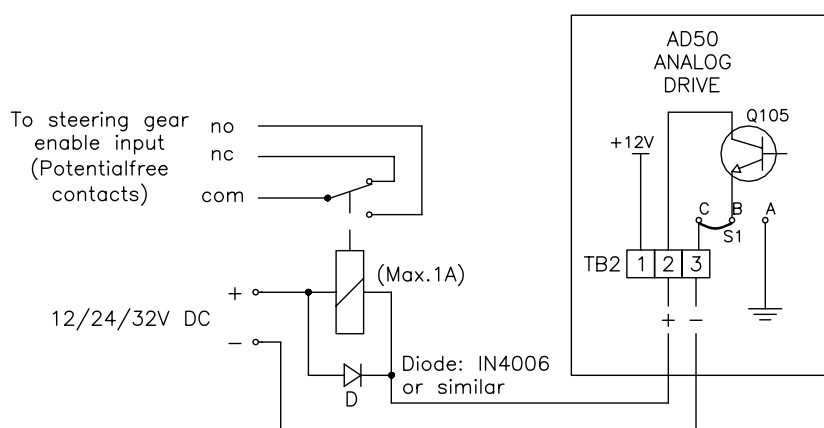


Figure 3-11 Rudder Enable, Alternative 2

Software Setup

When an AD50 and a rudder feedback unit are installed, the AP50 system automatically detects the rudder feedback unit during power up and makes the proportional rudder control function available. When there is an AD50 unit but no rudder feedback unit, analog control is available.

For proportional control, the continuous control voltage gives rudder speed to the steering gear. For analog control, the continuous control voltage gives rudder angle to the steering gear electronics.

Rudder direction and maximum and minimum values are set under the AP50 Installation menu. These settings determine the operating voltage range of the rudder control signal.

For software setup procedure, refer to the AP50 manual.

Rudder calibration

For proportional control, the minimum and maximum voltage for correct rudder speed is set automatically during the Dockside rudder test. Refer to *Rudder Test* under the Dockside menu in the AP50 Manual.

For analog control the voltage for maximum rudder travel and the corresponding angle is set individually for each direction. Refer to *Rudder Calibration* under the Dockside menu in the AP50 Manual.

Set Rudder Zero

For proportional control, the rudder zero is set during Seatrial when the boat steers straight ahead.

For analog control, an offset voltage can be applied in the Dockside menu for aligning AP50 rudder zero with zero angle for the ships main rudder angle indicators.

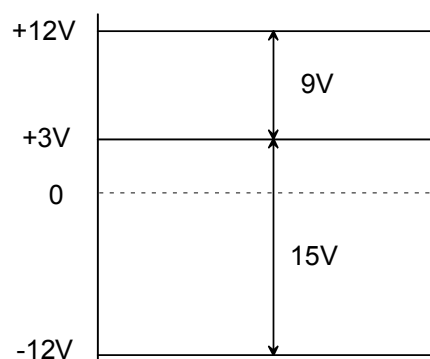
Default is half the available signal range.

Note !

Rudder Zero = 0 will give full range and equal amplitude of the signal to each side, but will be limited by the Maximum rudder setting. If Rudder Zero is set different from 0, part of the available signal range is used for permanent offset and the remaining range available for control will therefore be less in one direction.

Example:

Available range of $\pm 12V$ and Rudder Zero = $+3V$ gives $+9V$ and $-15V$ as remaining operational range. The $+9V$ and $-15V$ may if necessary be further reduced by Maximum Rudder setting.



4 MAINTENANCE

No special maintenance is required. However, it is advisable, to make a yearly check of all connections and visually inspect the inside of the unit.

4.1 Exchange of software program

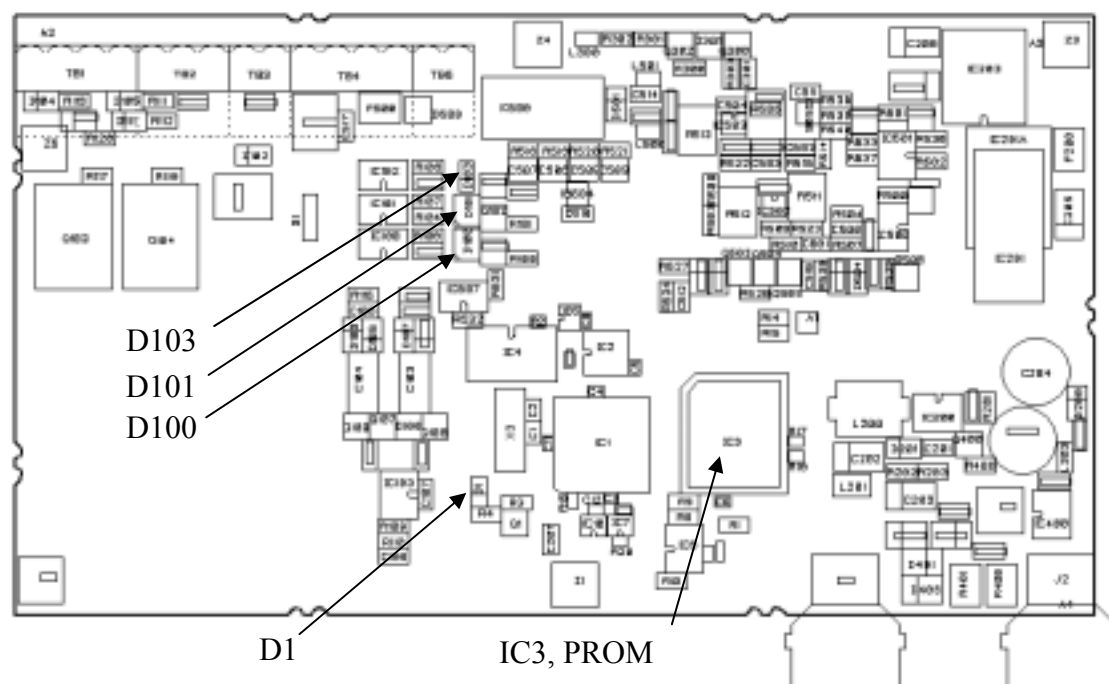
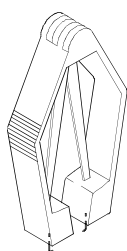
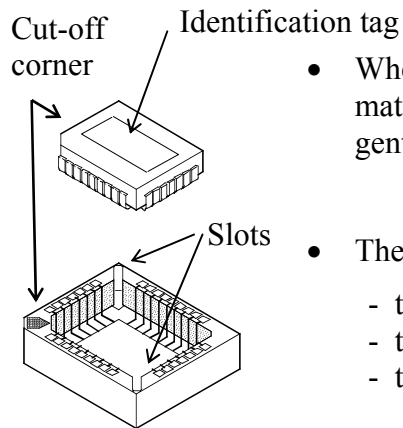


Figure 4-1 AD50 PCB, Component Layout



PROM extraction tool

- Remove the Programmable Read-Only Memory (PROM) from the socket with the PROM extraction tool (part number 44139806).
- Insert the tool by pressing the two grip pins down into the two slots in the corners of the socket.
- Squeeze the tool and pull out the PROM.



- When inserting new PROMS, make sure its cut-off corner matches with the corner in the socket. Press the PROM gently into the socket.
- The identification tag on the PROM indicates:
 - the name of the unit
 - the Simrad part number
 - the software version

Caution

Make sure that the correct PROM is mounted in each unit:

5 TROUBLESHOOTING

Audible and visual alarm is provided for every fault being detected. In the event of an autopilot failure, the AP50's numerous test features will assist you in isolating a probable fault. Refer to the AP50 Instruction Manual.

For analog rudder control, no rudder angle signal is available to the autopilot. Missing rudder response can therefore not be detected immediately and displayed. As described on page 20, this may result in "Vessel of course" alarm although the fault may be in the rudder control system.

In the unlikely event of a failure in the AD50, observe the operation of the different LED's according to the following table.

Also check that voltage is present on the input RobNet plug. Voltage shall be 26-40VDC.

Display readout	Probable fault	Recommended action
No rudder response (Remains in actual mode without any rudder command)	– Steering gear not operative – Broken connection – Missing power – Defective electronics	1. Check all connections. 2. Check that the AD50 is operative. The green LED D1 alternates with 1 Hz if the unit is OK. If the LED is permanently on, the processor has stopped. Switch off and on. If the LED is off, check power to the PCB or switch off and on. 3. If D1 alternates, check green LED D103 for Drive Enable (On in FU, AUTO and NFU). Also measure the continuous voltage (TB3 or TB4) and check the yellow LEDs D100 and D101 for port/stbd direction.

5.1 Failure Mode Effect Analyses

AD50 can support both analog and proportional rudder control. For analog control, a rudder angle voltage command is given but no rudder feedback is installed in the system. In proportional mode, a rudder speed voltage command is given and a rudder feedback unit gives rudder angle to the J50 Junction Unit. Should a technical fault occur, provisions have been made for safe operation according to the following table:

Cause of failure	Effect
<i>All modes</i>	
Power failure in autopilot system	Acoustic POWER FAILURE ALARM given (operated by separate alarm supply)
Processor stop or software lockup	Analog command line broken after approx. 1 sec. Watchdog restart after 1.6 seconds. “No rudder response” alarm after 5 sec.
Data line or power line broken in bus cable	After 5 sec: Analog command line broken Drive enable goes off “No rudder response” alarm
Ship does not respond to rudder command	“Vessel of course” alarm given when off course limit exceeded during transit
<i>Proportional mode (i.e. rudder feedback installed)</i>	
Rudder feedback signal lost	Rudder moved to commanded angle based on synthesized ¹⁾ rudder feedback. “Rudder feedback failure” given after 5 sec.
Steering gear failure, i.e. rudder feedback signal not changing when rudder command is given	“No rudder response” alarm after 5 sec. Rudder stop command given.

¹⁾ Rudder angle is calculated based on measured rudder speed and command time

6 SPARE PARTS LIST

20213088	AD50 Analog Drive
20212080	Installation Accessories
22086623	Box
20193264	Cover
44138816	Cover Nutknobs
20213096	AD50 Analog Interface PCB Ass'y
44139400	Cover for Plug
20213104	PROM (programmed) V..R..
20191607	RobNet Cable 7m (23') with Male Connectors

